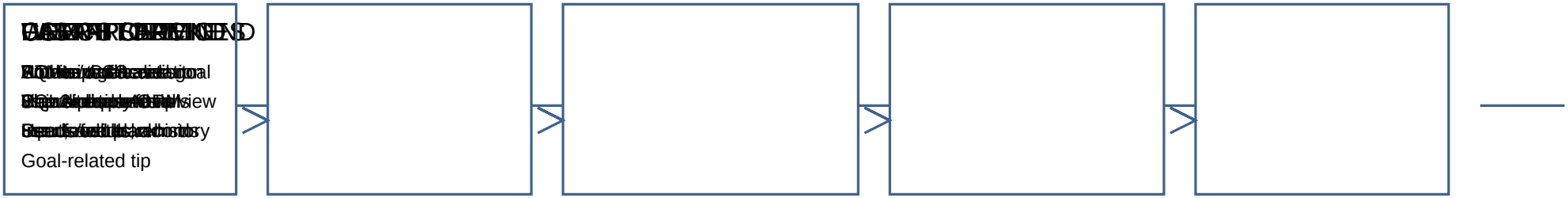
FitBuddy

Solution Architecture

Proposed logical architecture for the first release



COMPONENT RESPONSIBILITIES

Frontend: collects user input and displays the generated or revised plan.

FastAPI: validates requests, coordinates AI and database operations, and renders templates.

Gemini: creates workout plans, revises plans using feedback, and produces concise tips.

SQLite: stores user information, original plans, feedback, and updated plan versions.

Admin page: reads saved user and plan records for authorized coach/admin review.

KEY DESIGN NOTES

Keep API credentials on the server in environment configuration. Restrict admin routes.

Validate input and handle AI/database errors with clear user messages. Preserve original plans.

The diagram is conceptual; exact modules and model names depend on implementation choices.

Based on the supplied FitBuddy project brief.